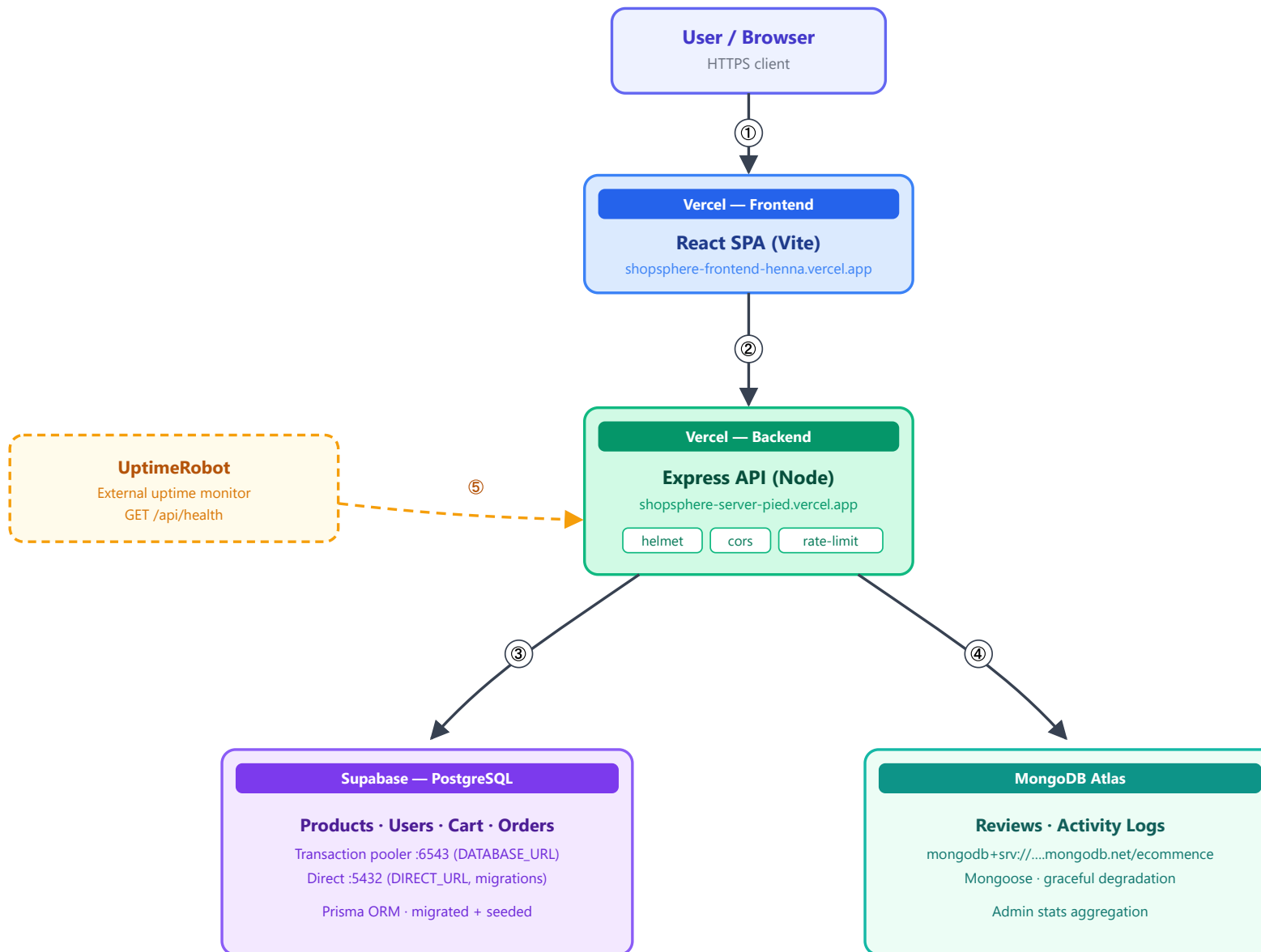


ShopSphere — Production Architecture (Deci Techno)

Task 1 production setup as deployed to Vercel + Supabase + MongoDB Atlas. Traffic routes shown with numbered edges.

Architecture Diagram



Cloud Service Classification

Service	Used as	Classification	Justification
Vercel Frontend shopsphere-frontend-henna.vercel.app	Hosting + CDN of the built React SPA	PaaS	Vercel builds and serves the static frontend with automatic TLS/CDN; we manage no servers, OS, or infrastructure.
Vercel Backend shopsphere-server-pied.vercel.app	Runs the Express API as serverless functions	PaaS	The API is deployed as managed Vercel serverless functions — Vercel handles runtime, scaling, and HTTPS; we only provide code and configuration.
Supabase	Managed PostgreSQL database	PaaS	A fully managed PostgreSQL platform (plus auth/APIs) connected over the Postgres wire protocol; we administer neither servers nor the database engine itself.
MongoDB Atlas (in use)	Managed MongoDB for reviews & activity logs	PaaS / DBaaS	Consumed as a hosted, managed MongoDB service via connection string — no server provisioning or maintenance.

Security & Delivery Notes

- **Zero hardcoded secrets:** all credentials (DATABASE_URL , DIRECT_URL , MONGODB_URI , JWT_SECRET) are injected via Vercel environment variables — none are committed to the repository.
- **Active middlewares:** helmet , strict cors (frontend origin only), and express-rate-limit (300 req / 15 min) are live on the deployed backend.
- **Health/uptime:** GET /api/health returns 200 only when PostgreSQL is reachable; monitored by UptimeRobot.
- **No local DB traffic:** the deployed app reads/writes strictly to Supabase (Postgres) and MongoDB Atlas.